Cryptography

Chapter 8

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Learning Objective

- Explain the basic principles of cryptography
- Describe the operating principles of the most popular cryptographic tools
- Cipher methods(Substitution/Transposition/Exclusive-OR)
- Cryptography Algorithm(Symmetric/ Asymmetric)
- List and explicate the major protocols used for secure communications
- Discuss the nature and execution of the dominant methods of attack used against cryptosystem.

Cryptography

- Cryptology, encompasses Cryptography and Cryptanalysis.
- Cryptography, which comes from the Greek words kryptos, meaning “hidden,” and graphein, meaning “to write,” is the process of making and using codes to secure the transmission of information.
- Cryptanalysis is the process of obtaining the original message (called the plaintext) from an encrypted message (called the ciphertext) without knowing the algorithms and keys used to perform the encryption.

Cryptography

- Encryption is the process of converting an original message into a form that is unreadable to unauthorized individuals—that is, to anyone without the tools to convert the encrypted message back to its original format.
- Decryption is the process of converting the ciphertext message back into plaintext so that it can be readily understood.

Terminology

- Algorithm: The programmatic steps used to convert an unencrypted message into an encrypted sequence of bits that represent the message;
- Cipher text or cryptogram: The encoded message resulting from an encryption process.
- Decipher: To decrypt, decode, or convert, cipher text into the equivalent plain text.
- Encipher: To encrypt, encode, or convert, plaintext into the equivalent cipher text.
- Key or crypto-variable: The information used in conjunction with an algorithm to create the ciphertext from the plaintext or derive the plaintext from the ciphertext;
- the key can be a series of bits used by a computer program,
- or it can be a passphrase used by humans that is then converted into a series of bits used by a computer program.

Terminology

- Keyspace: The entire range of values that can be used to construct an individual key.
- Link encryption: A series of encryptions and decryptions between a number of systems, wherein each system in a network decrypts the message sent to it and then re-encrypts it using different keys and sends it to the next neighbor, and this process continues until the message reaches the final destination.
- Plaintext or cleartext: The original unencrypted message, or a message that has been successfully decrypt.
- Steganography: The hiding of messages—for example, within the digital encoding of a picture or graphic.
- Work factor: The amount of effort (usually in hours) required to perform cryptanalysis to decode an encrypted message when the key or algorithm (or both) are unknown

Cipher Method

- There are two methods of encrypting plaintext:

1. Bit stream method
2. Block cipher method.

Cipher Method

Bit stream method :

- each bit in the plaintext is transformed into a cipher bit one bit at a time.
- Bit stream methods commonly use algorithm functions like the exclusive OR operation (XOR).

Cipher Method

- Block Cipher Method:
- In the block cipher method, the message is divided into blocks, for example, sets of 8-, 16-, 32-, or 64-bit blocks, and then each block of plaintext bits is transformed into an encrypted block of cipher bits using an algorithm and a key.
- Block methods can use substitution, transposition, XOR, or some combination of these operations.

Note that most computer-based encryption methods operate on data at the level of its binary digits (bits), but some operate at the byte or character level.

Substitution Cipher

- Substitute one value for another.
- For example a letter in the alphabet with the letter three values to the right.
- Or substitute one bit for another bit that is four places to its left.
- This is a simple enough method by itself but very powerful if combined with other operations.

Substitution Cipher

- Example:
- A three-character substitution to the right results in the following transformation of the standard English alphabet:

Initial alphabet yields	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Encryption alphabet	DEFGHIJKLMNOPQRSTUVWXYZABC

- Example:
- Within this substitution scheme,

the plaintext MOM would be encrypted into the ciphertext PRP.

Substitution Cipher

- With reference to previous example this type of substitution is based on a monoalphabetic substitution, because it only uses one alphabet.
- More advanced substitution ciphers use two or more alphabets, and are referred to as polyalphabetic substitutions.

Substitution Cipher

- Polyalphabetic substitutions:

Plaintext =	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Substitution cipher 1 =	DEFGHIJKLMNOPQRSTUVWXYZABC
Substitution cipher 2 =	GHIJKLMNOPQRSTUVWXYZABCDEF
Substitution cipher 3 =	JKLMNOPQRSTUVWXYZABCDEFGHI
Substitution cipher 4 =	MNOPQRSTUVWXYZABCDEFGHIJKL

- The first row here is the plain text ,and the next four rows are four sets of substitution ciphers, which taken together constitute a single polyalphabetic substitution cipher.

Substitution Cipher

Plaintext =	ABCDEFGHIJKLMNOPQRSTUVWXYZ
Substitution cipher 1 =	DEFGHIJKLMNOPQRSTUVWXYZABC
Substitution cipher 2 =	GHIJKLMNOPQRSTUVWXYZABCDEF
Substitution cipher 3 =	JKLMNOPQRSTUVWXYZABCDEFGHI
Substitution cipher 4 =	MNOPQRSTUVWXYZABCDEFGHIJKL

- To encode the word TEXT with this cipher, substitute a letter from the second row for the first letter in TEXT, a letter from the third row for the second letter, and soon—a process that yields the cipher text WKGF.
- Plain text = TEXT
- Cipher text= WKGF

Caesar Cipher

- Another type of mono alphabetic substitution cypher.
- The **Caesar Cipher** is one of the simplest and oldest methods of encrypting messages, named after Julius Caesar, who reportedly used it to protect his military communications.
- Julius Caesar reportedly used a three-position shift to the right to encrypt the messages (so A became D,B became E,and soon),and thus this particular substitution cipher was given his name—the Caesar Cipher.
- This technique involves shifting the letters of the alphabet by a fixed number of places.

Caesar Cipher

- Here is an example of how to use the Caesar cipher to encrypt the message “HELLO” with a shift of 3:

1. Write down the plaintext message: HELLO
2. Choose a shift value. In this case, we will use a shift of 3.
3. Replace each letter in the plaintext message with the letter that is three positions to the right in the alphabet.

```
H becomes K (shift 3 from H)
E becomes H (shift 3 from E)
L becomes O (shift 3 from L)
L becomes O (shift 3 from L)
O becomes R (shift 3 from O)
```

Plain Text= HELLO

Cypher Text= KHOOR

Hash Function

- Hash functions are mathematical algorithms that generate a message summary or digest (sometimes called a fingerprint) to confirm the identity of a specific message and to confirm that there have not been any changes to the content.
- Hash Function do not create a ciphertext, but confirm message identity and integrity.
- Hash algorithms are public functions that create a hash value, also known as a message digest, by converting variable-length messages into a single fixed-length value.

Hash Function

- The message digest is a fingerprint of the author's message that is compared with the recipient's locally calculated hash of the same message. If both hashes are identical after transmission, the message has arrived without modification.

Cryptographic Algorithm

- Symmetric Algorithm
- Asymmetric Algorithm

Symmetric Algorithms

- Same secret key to encipher and decipher the message what is called private key encryption or symmetric encryption.
- Symmetric encryption methods use mathematical operations that can be programmed into extremely fast computing algorithms so that the encryption and decryption processes are executed quickly by even small computers.
- One of the challenges is that both the sender and the recipient must have the secret key.
- Also, if either copy of the key falls into the wrong hands, messages can be decrypted by others and the sender and intended receiver may not know the message was intercepted.

Symmetric Encryption

- **Data Encryption Standard (DES)**
- There are a number of popular symmetric encryption cryptosystems.
- One of the most widely known is the Data Encryption Standard (DES), which was developed by IBM and is based on the company's Lucifer algorithm.
- DES uses a 64-bit block size and a 56-bit key for encryption.
- DES was adopted by NIST in 1976 as a federal standard for encryption, after which it became widely employed in commercial applications.
- DES enjoyed increasing popularity for almost twenty years, until In 1998, a group called the Electronic Frontier Foundation (www.eff.org), using a specially designed computer, broke a DES key in less than three days (just over 56 hours, to be precise).

Symmetric Encryption

- Triple DES (3DES) was created to provide a level of security far beyond that of DES.
- The successor to 3DES is the Advanced Encryption Standard (AES).
- AES is a federal information processing standard (FIPS) that specifies a cryptographic algorithm used within the U.S. government to protect information in federal agencies.
- AES implements a block cipher called the Rijndael Block Cipher with a variable block length and a key length of 128, 192, or 256 bits.
- Experts estimate that the special computer used by the Electronic Frontier Foundation to crack DES within a couple of days would require approximately 4,698,864 quintillion years (4,698,864,000,000,000,000,000) to crack AES.

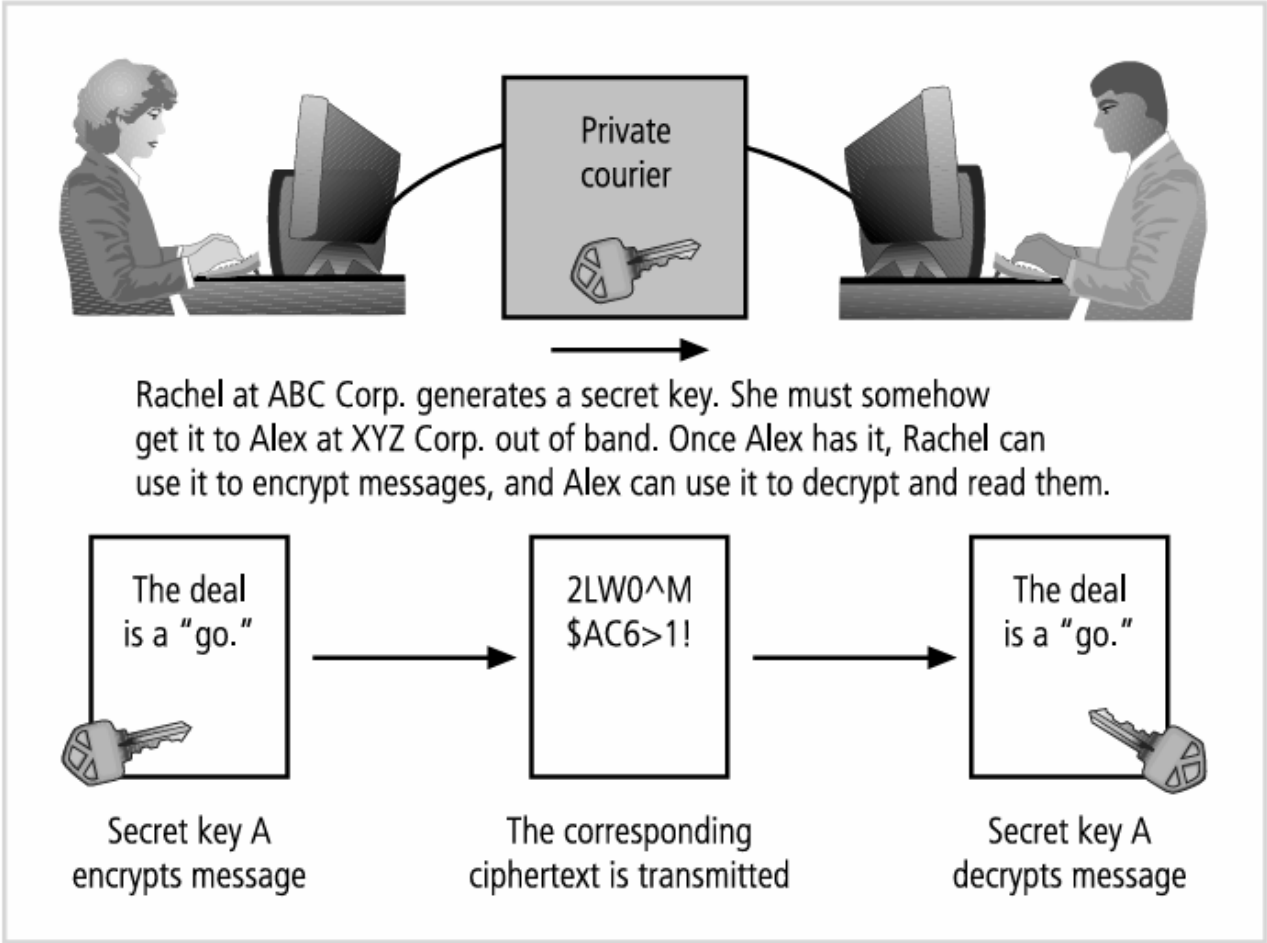
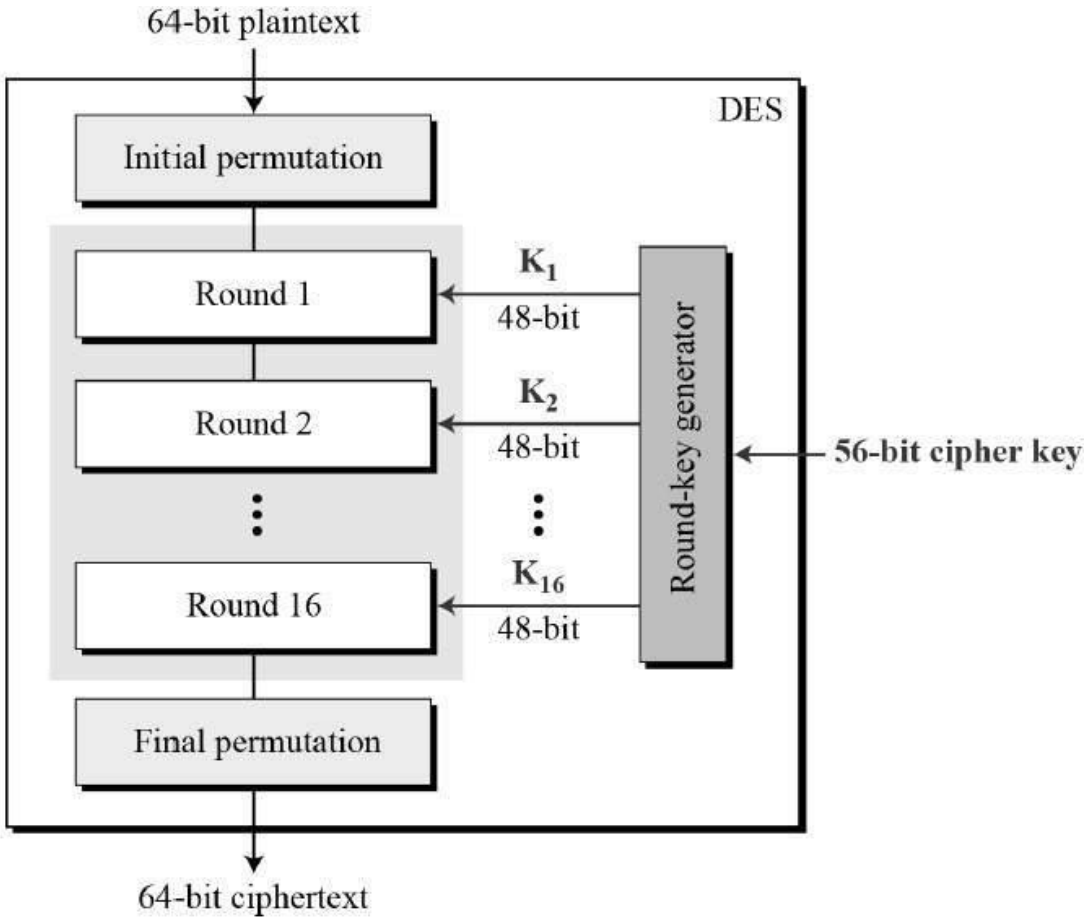
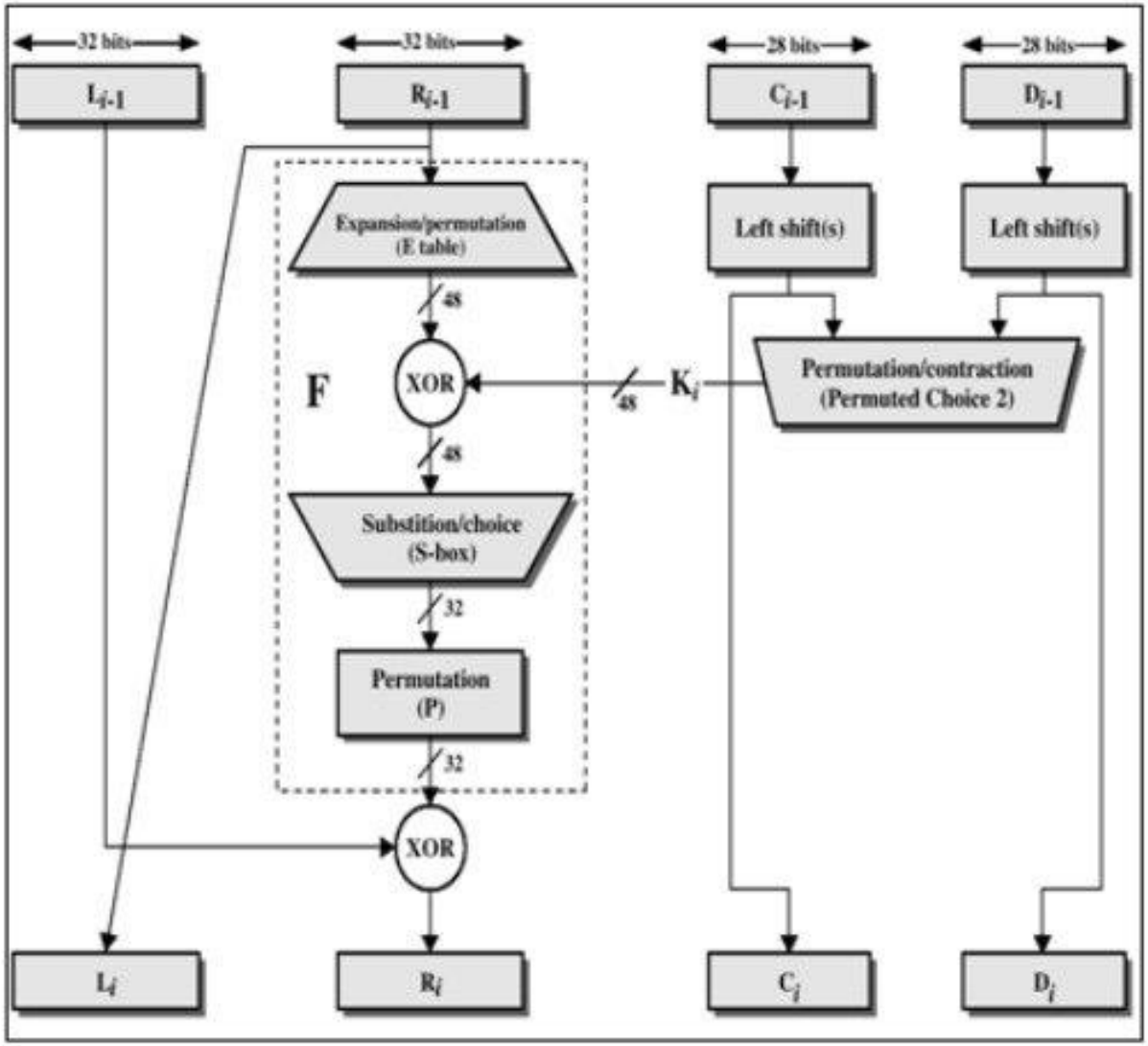


Figure 8-5 Example of Symmetric Encryption

DES Algorithm



DES Algorithm(each round)

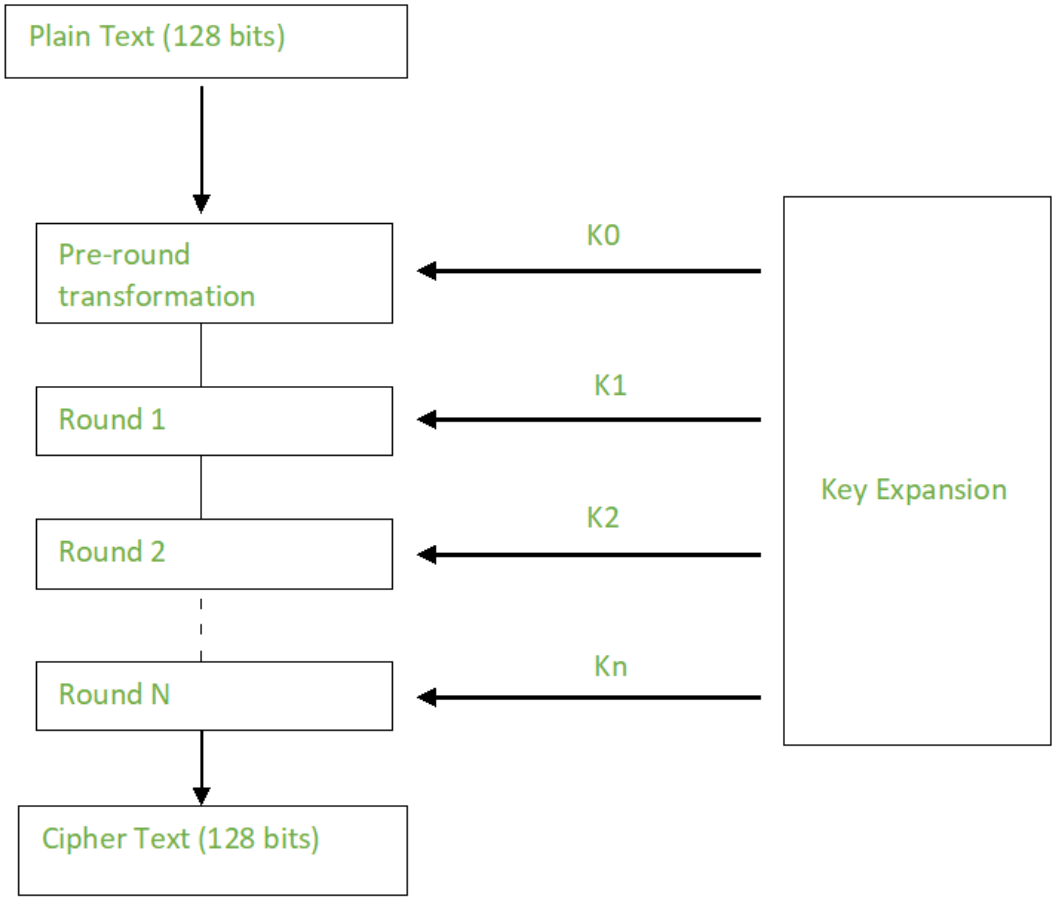


Advance Encryption Standard (AES) Algorithm

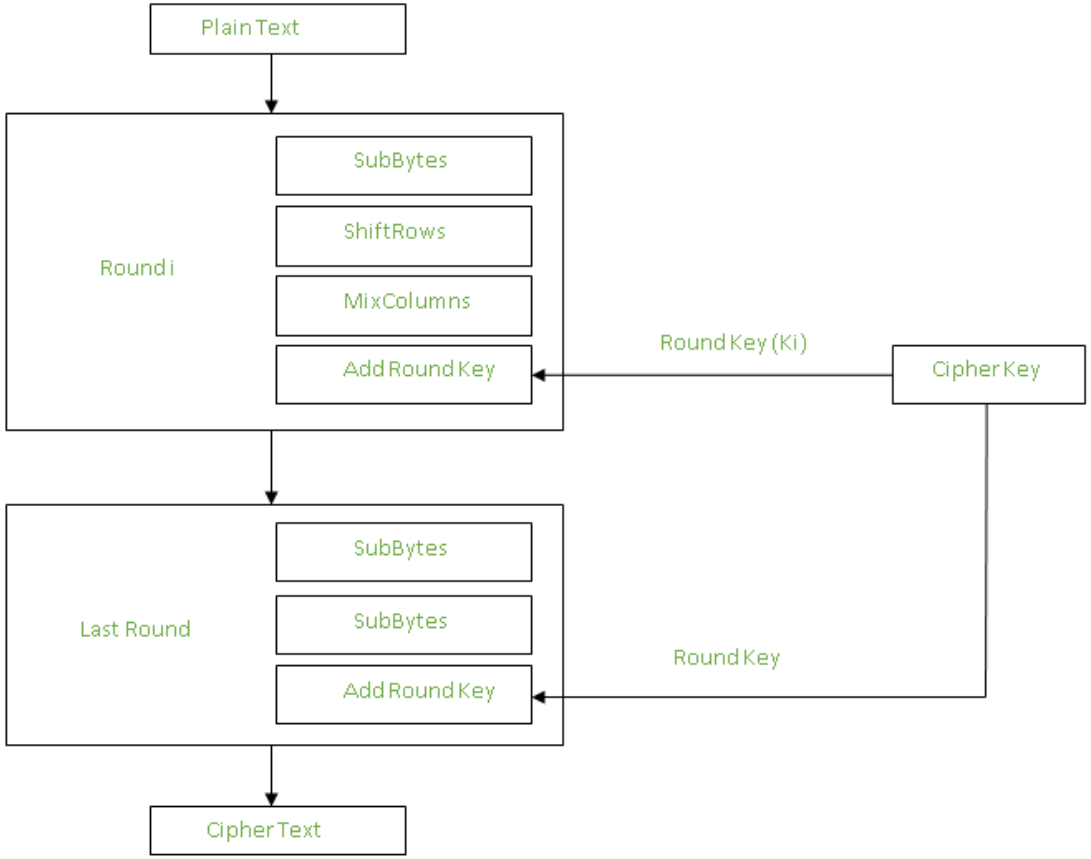
- Advanced Encryption Standard (AES) is a highly trusted **encryption algorithm** used to secure data by converting it into an unreadable format.
- Developed by the National Institute of Standards and Technology (NIST).
- **AES encryption** uses various **key lengths** (128, 192, or 256 bits) to provide strong protection against unauthorized access.
- **Points to Remember**
- AES is a Block Cipher.
- The key size can be 128/192/256 bits.
- Encrypts data in blocks of 128 bits each.

Advance Encryption Standard (AES) Algorithm

- 128 bits as input and outputs 128 bits of encrypted cipher text.
- AES relies on the substitution-permutation network principle, that is series of linked operations that involve replacing and shuffling the input data.



Creation of Round Keys (AES



Added Round Keys (AES)

AES Algorithm

- **Working:**

1. AES operates on bytes of data rather than in bits.
2. Block size is 128 bits, the cipher processes 128 bits (or 16 bytes) of the input data at a time.
3. The number of rounds depends on the key length as follows :
   1. 128-bit key – 10 rounds
   2. 192-bit key – 12 rounds
   3. 256-bit key – 14 rounds

Encryption

- AES considers each block as a 16-byte (4 byte x 4 byte = 128) grid in a column-major arrangement.

-
```
[ b0 | b4 | b8 | b12 |
| b1 | b5 | b9 | b13 |
| b2 | b6 | b10| b14 |
| b3 | b7 | b11| b15 ]
```

Each round comprises of 4 steps :

1. SubBytes
2. ShiftRows
3. MixColumns
4. Add Round Key

AES Algorithm

Sub Bytes

- **This step implements the substitution.**
- In this step, each byte is substituted by another byte.
- It is performed using a lookup table also called the S-box.
- This substitution is done in a way that a byte is never substituted by itself and also not substituted by another byte which is a compliment of the current byte. The result of this step is a 16-byte (4 x 4) matrix like before.
- The next two steps implement the permutation

AES Algorithm

- **Shift Rows**
- This step is just as it sounds. Each row is shifted a particular number of times.
- The first row is not shifted
- The second row is shifted once to the left.
- The third row is shifted twice to the left.
- The fourth row is shifted thrice to the left.

```
[ b0 | b1 | b2 | b3 ]      [ b0 | b1 | b2 | b3 ]
| b4 | b5 | b6 | b7 |  ->  | b5 | b6 | b7 | b4 |
| b8 | b9 | b10 | b11|     | b10 | b11 | b8 | b9|
[ b12 | b13 | b14| b15]    [ b15 | b12 | b13|b14]
```

AES Algorithm

- **Mix Columns**
- This step is a matrix multiplication. Each column is multiplied with a specific matrix and thus the position of each byte in the column is changed as a result.
- **This step is skipped in the last round.**

$$\begin{bmatrix} c0 \\ c1 \\ c2 \\ c3 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} b0 \\ b1 \\ b2 \\ b3 \end{bmatrix}$$

AES Algorithm

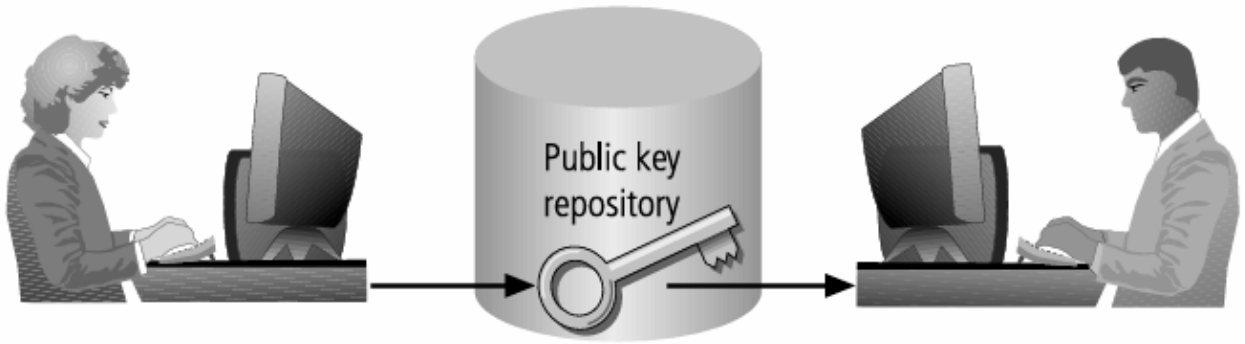
- **Add Round Keys**
- Now the resultant output of the previous stage is XOR-ed with the corresponding round key. Here, the 16 bytes are not considered as a grid but just as 128 bits of data.

Asymmetric Algorithms

- Asymmetric encryption uses two different but related keys, and either key can be used to encrypt or decrypt the message.
- If, however, key A is used to encrypt the message, only key B can decrypt it, and if key B is used to encrypt a message, only key A can decrypt it.
- One of the most popular public key cryptosystems is RSA, whose name is derived from Rivest-Shamir-Adleman, the algorithm's developers.
- The RSA algorithm was the first public key encryption algorithm developed (in 1977) and published for commercial use.

Asymmetric Algorithms

- It is very popular and has been embedded in both Microsoft and Netscape Web browsers to enable them to provide security for e-commerce applications.
- The problem with asymmetric is that holding a single conversation between two parties requires four keys.
- Moreover, if four organizations want to exchange communications, each party must manage its private key and four public keys.
- In such scenarios, determining which public key is needed to encrypt a particular message can become a rather confusing problem, and with more organizations in the loop, the problem expands.



Alex at XYZ Corp. wants to send a message to Rachel at ABC Corp. Rachel stores her public key where it can be accessed by anyone. Alex retrieves Rachel's key and uses it to create ciphertext that can be decrypted only by Rachel's private key, which only she has. To respond, Rachel gets Alex's public key to encrypt her message.

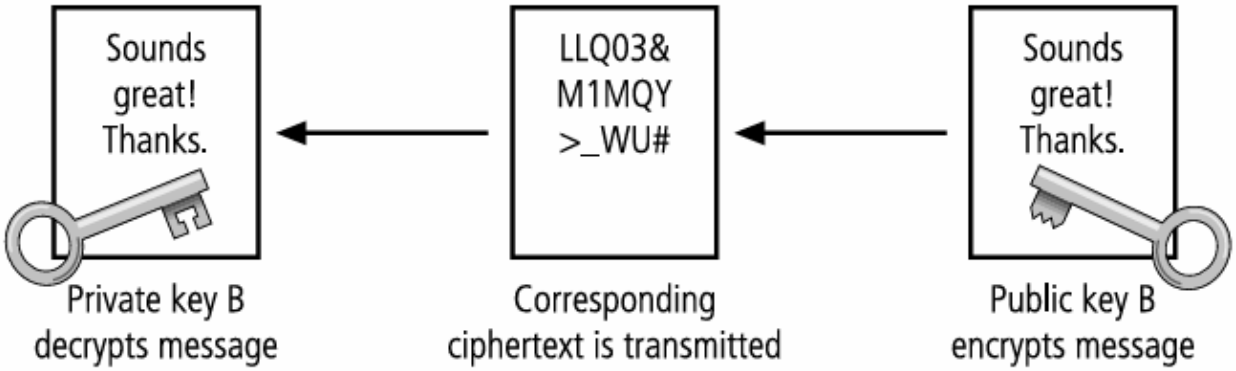


Figure 8-6 Example of Asymmetric Encryption

Questions